

Network System Implementation for the Marconi Law Firm, LLC.

Orlando, Florida

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Project Scope

MR. Marconi wanted help with implementing a new network for his new business space. Along with this he wanted to expand and grow his business further and wanted to have more capabilities.

The key objectives are to set up a network that is viable for the employees to use in a efficient and safe manner. The objectives are things like setting up the server, router, and switch. Also delivering IP addresses and OS that are supportive of one another while allowing all employees access to the server.

The project deliverables that are needed are things like proper hardware meaning a server rack with compatible HDD drives. Something else that is a deliverable in this project is having laptops with an OS like windows 10 pro to use on their client computers. Finally they also need something like server 2019 so that they are able to manage their server and who has access to said server. Things that are included in this project are things like 10 computers, 11 phones, 1 tv, 1 server, 1 router, 1 switch, 2 WAPs, 1 printer, operating systems, and server hard drives for a raid 5 system.

Things that are not included in this project are things like cabling (cat5(6)) also MR. Marconi didn't specify the operating system for the laptops or hardware. Things also like security both physical and cybersecurity weren't mentioned in the project.

The constraints for this project are like many others. These constraints are things like time, cost, scope, illness, injury, compatibility with hardware and software, and we also have things like patches and updates.

The 1st key success factor is to get the scope of my project signed and approved by all parties involved.

The 2nd key factor is going to sticking to the timeline made and ensuring the timeframe is being met and hopefully having some extra time incase mishaps happen.

The 3rd key factor will be to analyze and create a solid risk management plan for things like injury or even things like having secondary plans if something doesn't necessarily work as initially designed. For example, if a WAP doesn't work as well in one of the spots it was originally intended to be used.

The 4th key success factor is ensuring things are completed and completed correctly at each point of implementing new systems before starting on the next thing.

The 5th key factor will be maintaining the budget and staying within scope and cost of what was set initially. Ensuring that the project doesn't go too far out of scope or cost is vital to project success.

The 6TH key factor is to ensure a proper plan of attack is in place when starting the project. For example, having a step by step of when you are going to do things and how. This will allow you to ensure you're staying within your time frame and allow you to defend against a lot of those constraints that were mentioned above.

This project is expected to take 4 weeks and cost around \$50K.

Network Objectives/Benefits

- Title of benefit 1 – The first benefit Marconi will gain from this network is a RAID 5 system that will back up his data and valuable information to protect against attacks or data loss. This system allows you to maintain and save data, so you have a backup in case something happens to your system. This makes it so that even if you lose data, you can easily go to your back ups and just bring back all that lost data.
- Title of benefit 2 – the second benefit to Marconi will be the internet speeds and capabilities of his network to perform more tasks proficiently.
- Title of benefit 3 – The third benefit that this network is going to provide is security through groups and files being more secure and delegating the privileges of use to those who need access.

IP Network Design Table

-	IP Address
Netmask	255.255.255.0
First IP	10.10.119.0
Last IP	10.10.119.255
Total Host	256
Network ID	10.10.119.0
Broadcast Address	10.10.119.255

Node	Hardware Name	Static IP address
1	Marconi Computer	10.10.119.1
2	Jones Computer	10.10.119.2
3	Stark Computer	10.10.119.3
4	Drums Computer	10.10.119.4
5	Shoemaker Computer	10.10.119.5
6	Brooks Computer	10.10.119.6
7	Long Computer	10.10.119.7
8	Adams Computer	10.10.119.8
9	Smith Computer	10.10.119.9
10	Schultz Computer	10.10.119.10
11	Marconi VoIP Phone	10.10.119.101
12	Jones VoIP Phone	10.10.119.102
13	Stark VoIP Phone	10.10.119.103
14	Drums VoIP Phone	10.10.119.104
15	Shoemaker VoIP Phone	10.10.119.105
16	Brooks VoIP Phone	10.10.119.106
17	Long VoIP Phone	10.10.119.107
18	Adams VoIP Phone	10.10.119.108
19	Smith VoIP Phone	10.10.119.109
20	Schultz VoIP Phone	10.10.119.110
21	Router	10.10.119.253
22	Server	10.10.119.252
23	Switch	10.10.119.251
24	WAP 1	10.10.119.250
25	WAP 2	10.10.119.249
26	Printer	10.10.119.200
27	Smart TV	10.10.119.201
28	Conference Phone	10.10.119.202

Network Configuration Process

User Accounts		Setup and Configuration Process
Dan Marconi Phil Jones Karen Stark Evelyn Schultz Bill Shoemaker Mike Drums Lisa Brooks Terry Long Samuel Adams Brian Smith		Step 1: Launch Windows Server 2019 VM Step 2: Log into server using admin password Step 3: Open Server Manager by going to Windows Start Menu and click on Server Manager Next step: go to tools Next step: click on and open computer manager Next step: go to local/users and groups Next step: go to users Next step: right click in the white space Next step: click new user in the menu Next step: type in User name: "whatever username" Next step: fill out full name and description Next step: create and confirm password Next step: uncheck must change password on next log in Next step: click create Next step: once you click create it will create it and stay open so you may create another. Next step: repeat steps for each employee. Next step: Windows "user making" Next step: once all employees have been made on the server. Switch to your windows machine. Next step: click on the windows icon on the bottom left Next step: right click on your "user" Next step: click on edit settings Next step: click on family and other users Next step: add someone else to this PC Next step: click I don't have this info Next step: click as a user without Microsoft account Next step: fill out the info Next step: click next Next step: repeat for each user Next step: logout and log back in as the users to finalize each user. Next step: when logged in on windows machine in each user go to the search bar and type in your server path. Next step: for the network credentials type in that users info Next step: check remember me click Ok. Etc...
Groups		Setup and Configuration Process

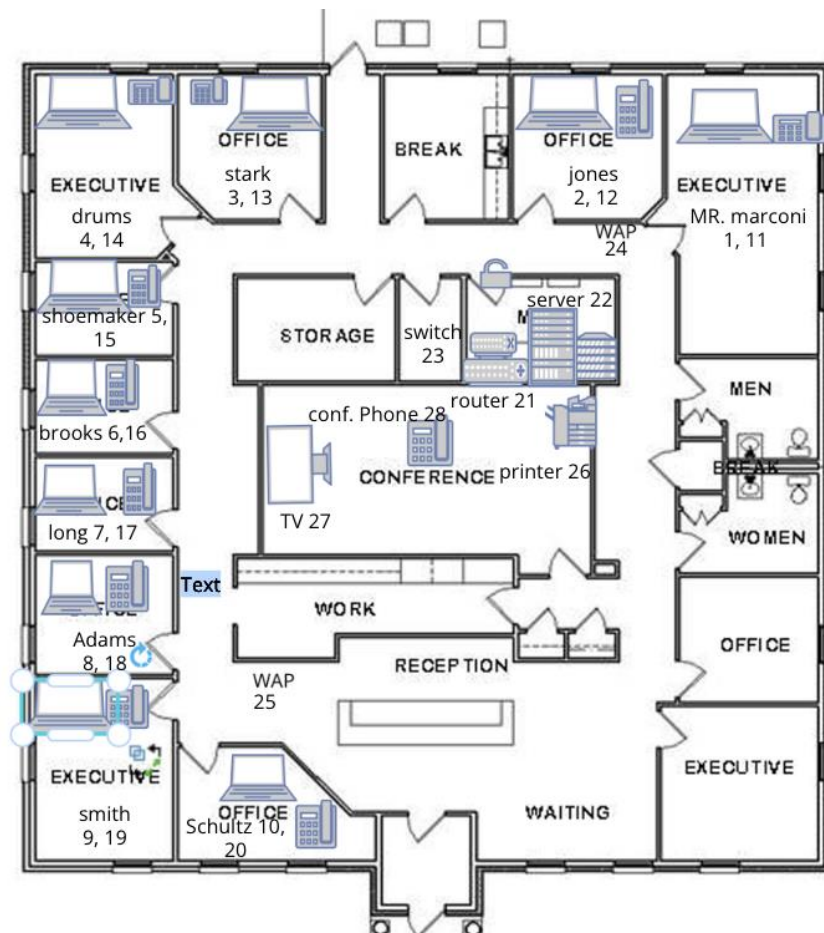
_Attorneys _Accounting _Administrative	Step 1: Launch Windows Server 2019 VM Step 2: Log into server using admin password Step 3: Open Server Manager by going to Windows Start Menu and click on Server Manager Next step: go to computer manager Next step: go to local/users and groups Next step: click groups Next step: right click on the white space Next step: select new group from the menu Next step: type group name “_<yourgroupname>” Next step: Members= user accounts Next step: click add button to add users Next step: add users that are part of that group (check names will double check name, to add multiple names you need to add a : between each name) Next step: click create Next step: repeat steps for each group Etc...
Folders and Permissions	Setup and Configuration Process

Attorneys Accounting Administrative	<u>Create Folders</u> Step 1: Launch File Manager Step 2: Double-click on Local C drive Next step 3: right click in the white space in the folder Next step: name folder "<main folder name>" Next step: enter the new folder you just created. Next step: repeat step 3 Next step: name new folder appropriate name Next step: repeat last 2 steps for each group (folder) Etc... <u>Set Sharing Permissions</u> Step 1: Launch Windows Server 2019 VM Step 2: Log into server using admin password Step 3: Open Server Manager by going to Windows Start Menu and click on Server Manager Next step: go to tools Next step: go to computer manager Next step: navigate to shared folders Next step: shared Next step: right click in white space Next step: new share Next step: click brows Next step: find folder you want Next step: click next, next, then click and check custom permissions. Next step: under share permission tabs (who had permission) Next step: remove "everybody" Next step: click add Next step: type in group you want to share permissions to for that certain folder Next step: give the group read and change access by selecting boxes (don't give full control) Next step: next click OK Next step: click finish Next step: repeat steps for each group/permission. Etc...
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User Logins and Passwords

Employee Name	Employee Title	Login ID	Password
Dan Marconi	Owner (log in) President/attorney	Dan Marconi	DMowner01*
Phil Jones	Attorney	Phil Jones	PJemployee02*
Karen Stark	attorney	Karen Stark	KSemployee03*
Evelyn Schultz	(log in) admin assist	Evelyn Schultz	ESemployee04*
Bill Shoemaker	attorney	Bill Shoemaker	BSemployee05*
Mike Drums	attorney	Mike Drums	MDemployee06*
Lisa Brooks	attorney	Lisa Brooks	LBemployee07*
Terry Long	attorney	Terry Long	TLemployee08*
Samuel Adams	attorney	Samuel Adams	SAemployee09*
Brian Smith	Accountant (log in)	Brian Smith	BSemployee10*

Network Topology Diagram with Employee Names and Node Numbers



Capital and Operating Costs

Hardware and Software	Per Unit Cost	Quantity	Total Cost
Laptops	1,253.99	10	12,539.9
Router	268.99	1	268.99
Server	7919.99	1	7919.99
Switch	849.98	1	849.98
Server Hard Drives (Consider a RAID 5)	577.99	4	2311.96
VoIP Phones	55.00	10	550.00
Conference Phone	185.99	1	185.99
WAPs	433.49	2	866.98
Printer	449.99	1	449.99
Smart TV	499.99	1	499.99
Windows Server 2019	1,639.99	1	1,639.99
Windows 10 Pro	Included w/laptop	0	0
Labor	\$125	40 hrs	\$5,000
Annual ongoing network maintenance and support	\$15,000	1 yr	\$15,000
GRAND TOTAL	-	-	\$48,083.76

Hardware and Software Specification Tables

Laptops

Brand/Manufacturer	Microsoft
Product Line	Microsoft surface Laptop GO 3
Model	Go 3
Processor/Clock Speed	Intel core I5 1235U /up to 4.4 GHz
Memory/Speed	16 GB Ram :PDDR5 SDRAM
Hard Drive/Capacity	512 GB SSD
Embedded Security	Microsoft defender Antivirus
Display/Size/Resolution	12.4 touchscreen 1536 x 1024
Networking	Wifi – 6, Bluetooth 5.1, IEEE 802.11a, IEEE 802.11ac, IEEE 802.11ax (Wi-Fi 6), IEEE 802.11b, IEEE 802.11g, IEEE 802.11n
Operating System	Win 10 PRO
Cache	12 megabyte
Page URL	https://www.cdw.com/product/microsoft-surface-laptop-go-3-12.4-core-i5-16-gb-ram-512-gb-ssd/7596457?pfm=srh#TS

Server

Brand/Manufacturer	HPE ProLiant
Product Line	ProLiant
Model	DL380 Gen10
Hard Drive/Capacity	N/A
Processor	Xeon Gold
Memory/Speed	Max ram 1.5 TB/ 2933 megahertz
Storage Controller	HPE Smart Array S100i SR Software
Monitor	
Ethernet	2 x 10 gigabit ethernet
Operating System	OS certified
Networking	10 gigabit ethernet
Expansion Bays	8 (total) / 8 (free) x hot-swap 2.5" SFF
Expansion Slots	1 (total) / 0 (free) x FlexibleLOM, 1 (total) / 1 (free) x microSD Card (internal), 1 (total) / 1 (free) x PCIe 3.0 x16 - full-length, full-height, 1 (total) / 1 (free) x PCIe 3.0 x8 - full-length, full-height, 1 (total) / 1 (free) x PCIe 3.0 x8 - half-length, full-height
Page URL	https://www.cdw.com/product/hpe-proliant-dl380-gen10-smb-networking-choice-rack-mountable-xeon-gold/6424503#TS

Server Hard Drives

Brand/Manufacturer	HPE
Product Line	Mixed Use
Storage Type	SSD
Hard Drive Type	Hot-swap hard drive
Form Factor	6.4 cm
Interface	Serial ATA-600
Capacity	960 GB
Data Transfer Rate (DTR)	600 MBps
Page URL	https://www.cdw.com/product/hpe-mixed-use-ssd-960-gb-sata-6gb-s/6424471#TS

Router

Brand/Manufacturer	Netgear
Product Line	Nighthawk AX6
Model	RAX50
RAM	512 megabyte
Flash Memory	256MB
WAN Ports Qty	1
Integrated Switch	4 LAN ports
Data Link Protocol	5 Gigbit Ethernet
Network/Transfer Protocol	TCP/IP, UDP, DHCP, DNS, HTTP/HTTPS, FTP, SMB/CIFS, NAT, UPnP, VPN (PPTP, L2TP, IPSec), QoS, SNMP
Page URL	https://www.cdw.com/product/netgear-nighthawk-ax6-rax50-wi-fi-6-ieee-802.11ax-ethernet-wireless-router/6180768?pfm=srh#TS

Switch

Brand/Manufacturer	Aruba
Product Line	HPE Aruba Instant On
Model	1930 48G Class4 PoE 4sFP/SFP+ 370W switch
RAM	512 megabytes
Flash Memory	256 megabytes
Performance	800 megahertz
Capacity	Access Control Entries (ACEs) per ACL: 480, IPv4 routes (static): 32, Link aggregation ports: 8, Packet buffer size: 1.5 MB, VLANs supported: 256
Ports Qty*	48
Page URL	https://www.cdw.com/product/hpe-networking-instant-on-1930-48g-class4-poe-4sfp-sfp-370w-switch-switc/7299319?pfm=srh#TS

Printer

Brand/Manufacturer	HP
Product Line	OfficeJet Pro
Model	9130b All-in-One Printer
Duty Capacity	30,000
Printing Output Type	color
Processor	1.2 GHZ
RAM	512 MB
Networking	Supported Network Protocols configurations (IPv4, IPv6); DHCPv4, DHCPv6; DHCP options: 81/RFC4702/RFC4704, 12-Hostname; 44; SMTP client; LLMNR; Bonjour/mDNS; LP/APIPA (Auto IP); NetBIOS/WINS; LPR/LPD; Port9100; DNS Resolver; SNMPv1/v2; Web Services Discovery; Web Services Print; Syslog; Firewall; SSL/TLS (HTTPS); Password-protected network Embedded Web server; Enable/disable network ports & device features, 802.1x wireless authentication (EAP-TLS, LEAP, and PEAP) & 802.1 Wired

Page URL	https://www.cdw.com/product/hp-officejet-pro-9130b-all-in-one-printer/7763529?pfm=srh#TS
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Wireless Access Points (WAPs)

Brand/Manufacturer	Linksys
Product Line	Linksys
Model	LAPAX3600C
Antenna	4x4:4 internal antennas for AX3600 Speeds (1200Mbps +2400Mbps)
Capacity	SSIDs total: 8
Networking	Maximum data rate (2.4 GHz): 1200 Mbps, Maximum data rate (5 GHz): 2400 Mbps (802.11a/b/g/n/ac/ax)
Interface	1 x 1 gigabit ethernet (high availability) – RJ-45, 1 x 100m/1g/2.5 Gigabit Ethernet – RJ-45
Power	POE+, AC 12 volt
Page URL	https://www.cdw.com/product/linksys-cloud-managed-ax3600-indoor-wireless-access-point/6736911?pfm=srh

Smart TV

Brand/Manufacturer	Samsung
Model	UN55TU8300F
Series	TU8300
Video Interface	54.6: viewable LED-backlit LCD TV – 4K
HDMI Ports Qty	3
Diagonal Size	55”
Resolution	3840 x 2160
Display Format	4k UHD (2160p)
Viewing Angle	16:9
Digital Audio Format	variety
USB Ports Qty	2
Wi-Fi Protocol	802.11ac
Page URL	https://www.cdw.com/product/samsung-un55tu8300f-tu8300-series-55-class-54.6-viewable-led-backlit/6292176?pfm=srh&expand=TS#TS

Conference Phone

Brand/Manufacturer	CISCO
Product Line	Cisco IP Phone
Model	7841
Type	VOIP
Call Services	Call Forwarding, Call Hold, Call Transfer, Call Waiting, Caller ID, Message Waiting Capability, Voicemail
Speakerphone	yes
Features	Call timer
Phone connections	Headset jack
Page URL	https://www.cdw.com/product/cisco-ip-phone-7841-with-multiplatform-phone-firmware-voip-phone-taa/4093496?pfm=srh#TS

Server Software

Brand/Manufacturer	Microsoft
Product Line	Microsoft windows
Version	Server 2019 standard
Licensing	10 clients, 16 cores, license, 64-bit
Page URL	https://www.cdw.com/product/microsoft-windows-server-2019-standard-license-10-clients-16-cores/5456567?pfm=srh#TS

Client Software

Brand/Manufacturer	INCLUDED IN LAPTOPS
Product Line	WINDOWS 10 pro
Version	Windows 10 pro
Licensing	N/A
Page URL	N/A

VoIP Phones

Brand/Manufacturer	Grandstream
Product Line	Grandstream small business IP phone
Model	GXP1615
Type	VOIP
Conference Call Capability	3-way
Call Services	Call Forwarding, Call Hold, Call Transfer, Call Waiting
Speakerphone	yes
Additional Functions	LCD display, 500 names & numbers
Features	Acoustic echo cancellation (AEC), Adaptive jitter buffer (AJB), Automatic gain control (AGC), Comfort noise generation (CNG), Packet loss concealment (PLC), Voice activity detection (VAD)
VoIP Protocols	SIP
IP Address Assignment	DHCP, PPoE, Static
Network Protocols	ARP, DNS, HTTP, HTTPS, ICMP, IP, Link Layer Discovery Protocol - Media Endpoint Discovery (LLDP-MED), RTCP, RTP, SRTP, TCP, TFTP, TR-069, UDP
PoE Support	Yes
Voice Features	G.771a,G.711u, G.722, G.723, G.726, G.729ab, iLBC, Opus
Page URL	https://www.cdw.com/product/grandstream-small-business-ip-phone-gxp1615-voip-phone-3-way-call-capab/5721269?pfm=srh#TS

Timesheets

Milestone	Activity Description (be descriptive)	Start Time	End Time	Total Activity Time	Activity Date
Milestone 1	Making the physical topography of the lawyer firm.	1500	1530	30	June 3 RD 2024
Milestone 1	Looking for items and filling out the bar graph	1500	1600	1HR	June 4 TH 2024
Milestone 1	Researching each piece of equipment/time table	1630	2000	4HR	June 5 TH 2024
Milestone 2	Making the IP list	1630	1700	30min	Jun 10 th 2024
Milestone 2	Made corrections and changes to topology	1600	1645	45 min	Jun 11 TH 2024
Milestone 2	Assigned Ips to different hardware	1700	1730	30 min	Jun 12 TH 2024
Milestone 3	Creating the users on server-2019	1600	1700	1HR	Jun 19 th 2024
Milestone 3	Added users to groups	1700	1730	30 mins	Jun 19 TH 2024
Milestone 3	Shared files between server 2019 and windows	1600	1730	1HR 30mn	Jun 20 TH 2024
Milestone 4					
Milestone 4					
Milestone 4					
TOTAL HOURS					

Status Reports

Accomplished this week

Milestone 1: I figured out all the details on the innovation and equipment the law firm may need in their renovations.
Milestone 1: I planned out the design and layout of the firm and where the server, router, and switch would be located
Milestone 1: I researched and compiled all the hardware and specs into bar graphs.
Milestone 2: This milestone allowed me to accomplish setting up a network ID, and a broadcast address.
Milestone 2: I planned out and designated. Static Ips to each individual device while also leaving room within the ranges to add new devices if needed.
Milestone 2: I planned for the next week by naming my hardware on server 2019 and windows 10.
Milestone 3: This milestone allowed me to see the difference between managing users on the server and on the windows system.
Milestone 3: I was able to plan out and configure users and their passwords.
Milestone 3: On top of the other two comments, I also shared and provided permissions to each user.
Milestone 4: I completed and adjusted the last of this report sheet.
Milestone 4: I completed creating and designing the slides for the.
Milestone 4: I completed presenting my project.

Planned for next week

Coming up for milestone 2: <i>for this week I plan to take whatever critics were made and see where I can improve and implement new ideas.</i>
Coming up for milestone 2: once understanding and knowing what is best changed, I will start a plan of attack on how to implement these changes.
Coming up for milestone 2: Once a plan of attack is in place I will research and find the hardware or gear that I need to complete the task.
Coming up for milestone 3: <i>I plan to take whatever critics were made and see where I can improve and implement or change some ideas.</i>
Coming up for milestone 3: I plan to take each user and plan a path to create and add users and privileges to the server so that the users may access it.
Coming up for milestone 3: after creating and setting up the users I will create and make their passwords.
Coming up for milestone 4: <i>This week I plan to take forward whatever feedback I may receive this week and put it to further use.</i>
Coming up for milestone 4: I also plan to get ready and provide a presentation of what will be done and how everything will be done. The “pitch”.
Coming up for milestone 4: I will look back on and reflect of everything that has gone on throughout the project and how I could take this into upcoming projects.

Comments for each milestone

Milestone 1:

What problems did you run into? A problem that I ran into is finding out exactly what exactly I needed to look for when regarding WAPs, Server racks, and windows server 2019. Finding some of the information was difficult as well. (in regard to specs of certain devices.)

How did you fix them? I did my diligence and did research to find the best possible use or what was used in similar situations. I also asked my professor and got his feedback on some of the issues.

Is there anything that you did not complete that you now must push off into the next milestone? Not that I ran into

Any other comments? I appreciate all the help this week!

Milestone 2:

What problems did you run into? This milestone was very refreshing when it came to re learning some of the things from NTE however an issue I ran into was how to best create static IP's while leaving room for growth and new similar devices so that they can share similar IP's.

How did you fix them? To fix this I grouped like things together like computers, phone, the backbone, and extra things like printers and gave them their IP range. For computers it was 1-100 for phones it was 100-200 and then for everything else it was 200 and above.

Is there anything that you did not complete that you now have to push off into the next milestone? No, I was able to complete all tasks.

Any other comments? No further comments on this milestone! Thank you for another period of instruction!

Milestone 3:

What problems did you run into? I ran into an issue with the permissions of the shared files for the accounting file. This issue was resolved after troubleshooting.

How did you fix them? I was able to trouble shoot and fix the issue by adding "users" to the permissions in the server manager shares screen. I also got further information provided in class. I ended up patching it again having to recreate the permissions through the computer manager and ensure permissions were the same as the other shared folders.

Is there anything that you did not complete that you now must push off into the next milestone? I was able to complete all of the objectives.

Any other comments? No further comments on this milestone! Thank you for another period of instruction!

Milestone 4:

What problems did you run into? During milestone 4 I didn't run into any issues.

How did you fix them? I fixed all the previous issues, so the presentation was easy.

Is there anything that you did not complete that you now must push off into the next milestone? I completed everything in this milestone.

Any other comments? N/A thanks for the knowledge and progressing me and my fellow peers further.